

Project Status Update: AI Research Collaboration with County of Santa Barbara's Zero Emission Vehicle Specialist

Project Overview

As part of the Professional Certificate in AI and Machine Learning at UC Berkeley, I (Carlos Muñoz Kampff) undertook a capstone project in collaboration with the County of Santa Barbara (COSB), working closely with Zero Emission Vehicle (ZEV) specialist Jerel Francisco. Over a seven-month period—including post-graduation—I engaged in biweekly consultations to support the County's sustainability initiatives through applied research in data science and artificial intelligence.

Research Question

How can the County of Santa Barbara's electric vehicle charging infrastructure data advance the implementation of the Zero Emission Vehicle Plan?

Research Objectives

The primary goal of the project was to explore how advanced data analysis, machine learning, and generative AI could enhance the County's ZEV Plan. Initial efforts focused on:

- Understanding COSB's existing data infrastructure and analytical tools for electric vehicle charging stations
- Reviewing relevant County literature and policy documents, including COSB's Climate Action Plan (CAP), Zero Emission Vehicle (ZEV) Plan (public draft), and the Countywide IT Strategic Plan
- Analyzing COSB's electric vehicle charging station utilization data
- Investigating demographic and statewide vehicle adoption trends using publicly available California Energy Commission (CEC) datasets

Methodology and Technical Contributions

The research process involved a combination of exploratory data analysis, predictive modeling, and prototype development. Key contributions included:

- **Statistical Analysis Reports:** Delivered comprehensive reports highlighting correlations, trends, and anomalies in COSB's charging station data to inform strategic planning.
- **Forecasting Models:** Developed STL-ARIMA and LSTM neural network models to predict future electricity demand for public charging infrastructure.
- **Scenario Evaluation:** Projected charging demand across multiple future scenarios using ZEV targets, CEC data (2010–2023), and empirical station usage data.
- **Decision Tree Modeling:** Developed decision tree models to uncover charging station utilization patterns, identifying a potential 27.3% efficiency gain. Analysis revealed that ~6% of civilian fleet sessions exceeded four hours, with a disproportionate number occurring near the County Jail. While discouraging extended sessions could improve grid performance, further data collection and stakeholder engagement are essential to understand underlying causes and avoid unintended impacts on marginalized communities.
- **Generative AI Dashboard:** Created a user-friendly dashboard that automatically generates EV charging station energy utilization forecasts, energy spike predictions and statistical summaries from uploaded datasets, translating complex outputs into accessible language.
- **Knowledge Base Interface:** Developed a generative AI-powered tool that allows users to query COSB documents—such as the CAP, ZEV draft plan, and IT Strategic Plan—using natural language.
- **Compliance and Governance Compatibility:** The prototypes in this project leverage open-source technologies including Next.js, Django, FAISS, Llama 3, TensorFlow, scikit-learn, MySQL, Pandas, and LangChain. As a result, the entire stack—including all AI and machine learning models—can be securely hosted within COSB's IT infrastructure without licensing fees or third-party dependencies as long as the required computational resources can be provisioned. Once fully developed, tested, and approved by COSB, it will be compatible with all foreseeable compliance requirements for deployment.

Outcomes and Recognition

The project culminated in presentations to both the UC Berkeley cohort and COSB stakeholders. It was reviewed by Professor Savio Saldanha and received UC Berkeley's "Exemplary Assignment Award" in recognition of its innovative methodology and practical impact on public sector sustainability planning.